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pHLIP®-mediated intracellular delivery of immuno-stimulatory compounds

Abstract

The invention features a composition comprising an immuno-stimulatory compound and a pHLIP® peptide, e.g., an immunostimulatory compound that comprises a cyclic purine dinucleotide, which binds to a stimulator of interferon genes (STING) such as a cGAMP cyclic compound inside a cell.

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Background/Summary

RELATED APPLICATIONS (1) This application is a national stage application filed under 35 U.S.C. § 371 of International Patent Application No. PCT/US2020/015402, filed Jan. 28, 2020, which claims the benefit of and priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 62/797,893, filed Jan. 28, 2019, the entire contents of which are incorporated herein by reference in its entirety.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING

(1) The contents of the sequence listing text file named "040984-511001WO_SL.txt." which was created on Jan. 28, 2020 and is 94,882 bytes in size, is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION

(2) The present invention relates to immunotherapy.

BACKGROUND

(3) Conventional methods of cancer treatment employ toxic agents that do not distinguish between normal and tumor tissues very well, and are therefore limited in their use because of their harmful side effects. Recent studies in cancer immunotherapy have shown that boosting the innate immune system of patients with cancer can have a critical effect in the outcome of cancer treatment, alone or in combination with conventional treatments Immuno-stimulating drugs exist, such as stimulator of interferon genes (STING) agonists, Toll-Like Receptor 7 (TLR-7) agonists, or retinoic acid-inducible gene I (RIG-I) agonists for instance, which have targets localized in the cytosol of a tumor cell. Although these drugs stimulate an immune response, it is necessary to not only specifically and efficiently target tumor cells to avoid adverse side effects but also to overcome the difficulty of introducing these drugs into the cytosol of tumor cells. Another limitation is that these drugs tend to be polar molecules that do not efficiently cross cell membranes. STING agonists, such as Cyclic guanosine monophosphate-adenosine monophosphate (cGAMP) for instance, are hydrophilic and negatively charged molecules. Such molecules are associated with poor membrane permeability, since cell membranes are non-polar, hydrophobic environments.

(4) Thus, a current challenge in the field of immunotherapy is to find methods to deliver these immune-stimulating drugs across cell membranes specifically into the cytosol of tumor cells, while avoiding delivery into healthy cells. Systemic delivery of STING agonists can be associated with off-target inflammation and/or a global autoimmune response which could be deleterious for the patient.

SUMMARY OF THE INVENTION

(5) The invention provides a solution to the problem of delivering STING and other immuno-stimulatory drugs to cancer cells and activated macrophages within the tumor microenvironment (TME). The compositions and methods described herein facilitate delivery of STING and other immuno-stimulatory drugs to cancer cells and macrophages, e.g., macrophages associated with tumors such as those within the TME.

(6) Tumors are characterized by a TME of a lower pH than the surrounding tissues, because of the metabolism accompanying their rapid and uncontrolled cell proliferation, which results in a flux of acidity emerging from tumor cells and/or macrophages associated with or infiltrating a tumor. Moreover, due to the flux and the membrane potential, the extracellular pH is lowest at the surfaces of cancer cells (meaning the pH is in close vicinity of the cellular membrane or the pH is measured just near the cell membrane) and is significantly lower than the bulk extracellular pH (the pH at the distance from cell membrane) in tumors compared to normal tissues of the same tissue type. The low pH region persists at the cancer cell surfaces even in well-perfused tumor areas.

(7) A pH Low Insertion Peptide (pHLIP®) is a water-soluble membrane peptide that interacts weakly with a cell membrane at neutral pH, without insertion into the lipid bilayer; however, at slightly acidic pH (<7.0), pHLIP® inserts into the cell membrane and forms a stable transmembrane alpha-helix. In addition to tumor cells characterized by low pH (<7.0), immune cells within a tumor mass are also characterized by low pH (<7.0). For example, the cells within the environment of a tumor mass, e.g., macrophages, are also characterized by low pH. By binding a pHLIP®, or pHLIP® equivalent, to an immune-stimulating drug, the compositions and methods specifically deliver the drug directly to the cancerous and immune (macrophages) cells and into their cytosols due to their acidic cell surfaces. Delivering immune-stimulatory compounds or drugs using pHLIP® peptides boosts the immune system, e.g., by stimulating the production of interferon-gamma, to mount and/or establish a tumor specific immune response and fight the cancer.

(8) A significant advantage of this approach is that the augmented immune response stimulated by the pHLIP® constructs described herein are associated with few to no side effects for the patient.

(9) Accordingly, the invention features a composition comprising an immuno-stimulatory compound and a pHLIP® peptide, e.g., an immunostimulatory compound that comprises a cyclic purine dinucleotide or its derivatives, which bind in a cell to STING. For example, the immunostimulatory compound binds to a STING inside a cell. Examples of STING compounds include Cyclic guanosine monophosphate-adenosine monophosphate (cyclic GMP-AMP (cGAMP)) or derivatives thereof, e.g., a —NH.sub.2 or —SH derivative of the STING compound, 3',5'-cyclic diadenylic acid (c-di-AMP), or a cyclic diguanylate (c-di-GMP) cyclic compound, or a derivative thereof. In some examples, the immuno-stimulatory compound is a polar compound or a moderately hydrophobic compound. Such compounds are characterized by limited intracellular penetration by passive diffusion across a plasma membrane. In other examples, the immuno-stimulatory compound is delivered into a cell. A tumor cell or immune cell comprising a composition comprising an immuno-stimulatory compound and a pHLIP® peptide is also within the invention.

(10) A compound is characterized as polar if it has a log P of less than -0.4. The immunostimulatory compound may be moderately hydrophobic. Exemplary cargo compounds, e.g., immune-stimulatory compounds or drugs, are polar, moderately hydrophobic or hydrophobic as defined by the following characteristics. Polar: Log P < -0.4; Moderately hydrophobic: 2.5 < Log

P < -0.4; and Hydrophobic: Log P > 2.5. The polarity and/or hydrophobicity of an immunostimulatory compound is measured using methods known in the art, e.g., by determining Log P, in which P is the octanol-water partition coefficient. A substance is dissolved into an octanol-water mixture, mixed, and allowed to come to equilibration. The amount of substance in each (or one) phases is then measured. The measurements itself could be in a number of ways known in the art, e.g., by measuring absorbance, or determining the amount using NMR, HPLC, or other known methods. As described herein, moderately hydrophobic, for example, is defined as molecule with Log P value in the range of 2.5 to -0.4, there are a lot of examples. In some examples, the immunostimulatory compound induces its biological effect only when it is inside a cell. For example, the immunostimulatory compound by itself has limited intracellular penetration by passive diffusion across the plasma membrane. The invention provides a solution to the problem of limited intracellular penetration to target cells (such as tumor cells and tumor-associated immune cells), because pHLIP® peptide sequences mediate targeting of the immune-stimulatory compound to tumor cells and/or immune cells such as macrophages in the TME, and subsequent delivery of the compound into the targeted cells. To be active, the immunostimulatory compound needs to be delivered into cells by pHLIP®, i.e., the immune-stimulatory compound cannot efficiently gain access to the cytosol without pHLIP®, which mediates its intracellular delivery.

(11) In some embodiments, the composition further comprises a linker between the immunostimulatory compound and the pHLIP® peptide. Exemplary linkers include a disulfide bond or an acid-labile bond. In some examples, the linker is cleavable. In other examples, the linker is not cleavable. Exemplary cleavable linkers include those that are self-immolating. The purpose of self-immolating linker is to restore a drug, e.g., an immune-stimulatory compound such as a STING agonist, to its original structure following cleavage of the linker that linked it to the pHLIP® peptide. Self-immolative elimination is a spontaneous and irreversible disassembly of a multicomponent compound into its constituent fragments through a cascade of electronic elimination processes. Self-immolative elimination is driven by an increase in entropy coupled with the irreversible formation of thermodynamically stable products (e.g. CO₂). Such linkers have an advantage in that the cargo/therapeutic agent (drug) can be released in an unmodified form if it has an appropriate —NH₂ or —OH group, such as

(12) ##STR00001##

(13) In general self-immolative linkers are well known linkers and widely used.

(14) A modulator of polarity is optionally included in the composition. Such a modulator changes the overall polarity of the construct to optimize delivery to cancer and immune cells or a tumor mass. If the cargo is polar (Log P < -0.4), the hydrophobic modulator will increase the Log P of [the cargo-modulator] (Log P > -0.4). If the cargo is hydrophobic Log P > 2.5, the polar modulator will decrease the Log P of [the cargo-modulator] (Log P < 2.5). Non-limiting examples of modulators are fatty acids, PEG polymers, hydrophobic fluorescent dyes, or cyclic peptides. For example, if the cargo renders the composition too polar, a modulator agent is added to make the overall composition less polar, or if the cargo is not polar enough, a modulator is added to make the composition more polar. For example, linkers comprising such modulators have an advantage in enhancing the efficiency of drug delivery into the cytosol or improving the targeting of tumors relative to normal tissues. In some examples, the construct may include a polar modulator; in other examples (such as in the case of a polar drug), the construct may include a more hydrophobic modulatory to promote delivery into the cell. As used herein, modulators are used for intracellular delivery of cytotoxic molecules.

(15) In some examples, the composition comprises 2 or more pHLIP® peptides. Exemplary constructs comprise the following structure: Peptide-L-B, in wherein “Peptide” is a first pHLIP® peptide comprising the sequence ADDQNPWRAYLDLLFPTDTLLLDLLWXA (SEQ ID NO: 1) or ADQDNPWRAYLDLLFPTDTLLLDLLWXA (SEQ ID NO: 214), “B” is a second pHLIP® peptide comprising the sequence ADDQNPWRAYLDLLFPTDTLLLDLLWXA (SEQ ID NO: 1)

or ADQDNPWRAYLDLLFPTDTLLLDLLWXA (SEQ ID NO: 214), wherein upper case “X” indicates any amino acid residue and can include lysine (Lys), cysteine (Cys), or an Azido-containing amino acid; and “L” is a polyethylene glycol linker, and each “—” is a covalent bond. (16) Thus in some compositions and methods, pHLIP® peptide comprises the amino acid sequence of Var3 pHLIP® sequence ADDQNPWRAYLDLLFPTDTLLLDLLWCA (SEQ ID NO: 10, ADQDNPWRAYLDLLFPTDTLLLDLLWCA (SEQ ID NO: 212), or variations thereof. In some examples, the pHLIP® peptide comprises Var3 sequence with the amino acid sequence of ADDQNPWRAYLDLLFPTDTLLLDLLWXA (SEQ ID NO: 215) or ADQDNPWRAYLDLLFPTDTLLLDLLWXA (SEQ ID NO: 216), where X is a functional group for conjugation purposes. For example, the functional group is an amino acid, e.g., an amino acid selected from lysine (Lys), cysteine (Cys), Azido-containing amino acid or others (e.g., any modified amino acid for conjugation purposes).

(17) Also within the invention is a method of augmenting an anti-tumor immune response, comprising administering to a subject a composition comprising an immuno-stimulatory compound and a pHLIP® peptide as described above. In some examples, the immuno-stimulatory compound is a polar compound such as a STING compound. For example, the subject comprises a solid tumor. The composition is administered using methods known in the art; e.g., the composition is injected directly into a tumor mass. Alternatively, the composition is systemically administered. The immuno-stimulatory compound is delivered into the cytosols of cancer cells and/or the immuno-stimulatory compound is delivered into the cytosols of macrophages within the tumor microenvironment. Because of the unique targeting aspect of the pHLIP® construct to tumors, systemic administration is possible—an important advantage of this therapy as such delivery is less invasive and traumatic to the subject to be treated.

(18) Certain implementations comprise a formulation for a parenteral, a local, or a systemic administration comprising a pHLIP®-linker-Cargo (e.g., an immuno-stimulatory compound).

(19) Formulations comprising a pHLIP®-linker-Cargo for intravenous, subcutaneous, intraarterial, intraperitoneal, intracerebral, intracerebroventricular, intrathecal, intracardiac, intracavernous, intraosseous, intraocular, or intravitreal administration are also provided.

(20) In an aspect, provided herein is a formulation comprising a pHLIP®-linker-Cargo for intramuscular, intradermal, transdermal, transmucosal, intralesional, subcutaneous, topical, epicutaneous, extra-amniotic, intravaginal, intravesical, nasal, or oral administration.

(21) The present subject matter also includes a formulation for intravesical instillation comprising a pHLIP®-linker-Cargo as disclosed herein. In some embodiments, the formulation is used for the treatment of cancer (e.g., solid tumors) or autoimmune diseases.

(22) Also provided herein is a formulation comprising a pHLIP®-linker-Cargo that comprises multiple pHLIP® peptides for systemic administration. In certain embodiments, the formulation is used for the treatment of cancer or autoimmune diseases or inflammation.

(23) Provided herein is a method of treating cancer or autoimmune diseases or inflammation in a subject, comprising administering to the subject an effective amount of a pH-triggered compound, wherein the compound comprises an immune-stimulating cargo compound. Non-limiting examples of cancer include colon cancer, prostate cancer, breast cancer, bladder cancer, lung cancer, skin cancer, liver cancer, bone cancer, ovarian cancer, stomach cancer, pancreatic cancer, testicular cancer, and brain cancer. In some embodiments, the cancer is bladder cancer. Also included herein are methods for detecting and/or imaging diseased tissue (such as cancer tissue) in a subject comprising administering to the subject with a pHLIP®-Linker-Cargo conjugated with imaging agent (I.A.), such as I.A.-pHLIP®-Linker-Cargo. Non-limiting examples of imaging agents include fluorescent dyes or nuclear imaging agents.

(24) The invention encompasses a method of augmenting an anti-tumor immune response, comprising administering to a subject a composition comprising an immuno-stimulatory compound and a pHLIP® peptide. For example, the subject comprises a solid tumor, e.g., cancer types

described above. The composition is injected directly into a tumor mass or is systemically administered. The immuno-stimulatory compound is delivered into the cytosols of cancer cells and tumor-activated macrophages by virtue of its presence in a composition that includes both the immune-stimulatory compound and pHLIP®. For example, the composition is delivered into the cytosol of a macrophage within the diseased tissue.

(25) Because of the presence of pHLIP® in the composition, the immuno-stimulatory compound is delivered intracellularly to induce a biological effect. For example, the biological effect of the immuno-stimulatory compound delivered in the presence of pHLIP®, e.g., in a composition that comprises both components, is at least 20%, 50%, 2-fold, 5-fold, or greater than the biological effect of the immune-stimulatory compound delivered in the absence of pHLIP® in the composition. In the absence of pHLIP®, the immuno-stimulatory compound comprises limited intracellular penetration by passive diffusion across a plasma membrane.

(26) The composition targets the immune-stimulatory compound preferentially to a diseased tissue compared to a healthy tissue, thereby minimizing damage to the healthy tissue. For example, the composition selectively promotes intracellular delivery of the immuno-stimulatory compound to cells in diseased tissue, e.g., the composition selectively promotes intracellular delivery of the immuno-stimulatory compound into a cancer cell.

(27) Macrophages within a tumor environment or the environment of a diseased tissue (e.g., inflamed tissue) are also acidic. Thus by virtue of the presence of pHLIP® in the composition, the composition selectively promotes intracellular delivery of the immuno-stimulatory compound into macrophages in a tumor microenvironment or into a macrophage in another diseased tissue environment. Delivery of an immune-stimulatory compound immuno-stimulatory compound delivered in the presence of pHLIP®, e.g., in a composition that comprises both components—immunostimulatory compound and pHLIP®, is at least 20%, 50%, 2-fold, 5-fold, or greater than the amount of the immuno-stimulatory compound delivered to tumor-associated macrophages in the absence of pHLIP® in the composition.

(28) Included herein are pharmaceutical compositions comprising a pH-triggered compound and a pharmaceutically acceptable carrier.

(29) As used herein, “effective” when referring to an amount of a compound refers to the quantity of the compound that is sufficient to yield a desired response. For example, the amount of cargo, e.g., immune-stimulatory compound yields a desired response, e.g., augmentation of an anti-tumor effect, without undue adverse side effects (such as toxicity, irritation, or allergic response) commensurate with a reasonable benefit/risk ratio when used in the manner of this disclosure.

(30) In some embodiments, a subject is a mammal. In certain embodiments, the mammal is a rodent (e.g., a mouse or a rat), a primate (e.g., a chimpanzee, a gorilla, a monkey, a gibbon, a baboon), a cow, a camel, a dog, a cat, a horse, a llama, a sheep, a goat, or a pig. In preferred embodiments, the subject is a human.

(31) Also within the invention is a cell, e.g., a tumor cell or a tumor-associated immune cell such as a macrophage comprising an immuno-stimulatory compound and a pHLIP® peptide as described above.

(32) Each embodiment disclosed herein is contemplated as being applicable to each of the other disclosed embodiments. Thus, all combinations of the various elements described herein are within the scope of the invention.

(33) Other features and advantages of the invention will be apparent from the following description of the preferred embodiments thereof, and from the claims. Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, suitable methods and materials are described below.

Description

DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram of a pHLIP® construct with an immune-stimulatory compound (ISC).
- (2) FIGS. 2A-2C are diagrams of pHLIP® constructs with an ISC and a modulator (M) molecule. The modulator could be attached to the pHLIP® peptide (FIG. 2A), linker (FIG. 2B) or ISC (FIG. 2C).
- (3) FIG. 3 is a diagram of a pHLIP® construct with 2 (or more) ISCs.
- (4) FIGS. 4A-4C are diagrams of pHLIP® constructs with 2 (or more) ISCs and a M molecule. The modulator could be attached to pHLIP® peptide (FIG. 4A), linker (FIG. 4B) or ISC (FIG. 4C).
- (5) FIG. 5 is a diagram of a pHLIP® construct with an ISC and an imaging agent at the membrane non-inserting end.
- (6) FIGS. 6A-C are diagrams of pHLIP® constructs with an ISC and M molecule and an imaging agent at the membrane non-inserting end. The modulator could be attached to pHLIP® peptide (FIG. 6A), linker (FIG. 6B) or ISC (FIG. 6C).
- (7) FIG. 7 is a diagram of a pHLIP® construct with multiple ISCs and an imaging agent at the membrane non-inserting end.
- (8) FIGS. 8A-C are diagrams of pHLIP® constructs with multiple ISCs and M molecule and an imaging agent at the membrane non-inserting end. The ISCs can be the same or different. The modulator could be attached to pHLIP® peptide (FIG. 8A), linker (FIG. 8B) or ISC (FIG. 8C).
- (9) FIG. 9 is a diagram of two or more pHLIP® peptides connect to each other by a PEG polymer (or any other polymer—shown by purple color) with an ISC linked together via a linker molecule.
- (10) FIGS. 10A-C are diagrams of exemplary pHLIP® constructs with two or more pHLIP® peptides connect to each other by PEG polymer (or any other polymer—shown by purple color) with an ISC linked together via a linker molecule. The modulator could be attached to pHLIP® peptide (FIG. 10A), linker (FIG. 10B) or ISC (FIG. 10C).
- (11) FIG. 11 is a diagram of an exemplary pHLIP® construct with two or more pHLIP® peptides connect to each other by PEG polymer (or any other polymer—shown by purple color) with an ISC linked together via a linker molecule.
- (12) FIGS. 12A-C are diagrams of exemplary pHLIP® constructs with two or more pHLIP® peptides connect to each other by PEG polymer (or any other polymer—shown by purple color) with an ISC linked together via a linker molecule. The modulator could be attached to pHLIP® peptide (FIG. 12A), linker (FIG. 12B) or ISC (FIG. 12C).
- (13) FIG. 13A is a structure of an exemplary self-immolating linker with immuno-stimulatory compound (ISC) for S-S exchange with Cys residue of pHLIP® peptide.
- (14) FIG. 13B is a structure of an exemplary self-immolating linker with immuno-stimulatory compound (ISC) for conjugation with Lys residue at pHLIP® peptide.
- (15) FIG. 14A is a chemical structure of c-GAMP (ADU-S100 (MIW815)) cyclic dinucleotide (CDN) GMP-AMP, agonist (activator) of STING.
- (16) FIG. 14B is a chemical structure of c-diAMP cyclic dinucleotide (CDN) di-AMP, agonist (activator) of STING.
- (17) FIG. 14C is a chemical structure of c-diGMP cyclic dinucleotide (CDN) di-GMP, agonist (activator) of STING.
- (18) FIG. 15A is a chemical structure of a pHLIP-S-S-cGAMP in which cGAMP is coupled with Cys at pHLIP peptide via self-immolating linker.
- (19) FIG. 15B is a chemical structure of a pHLIP-S-S-cGAMP in which cGAMP is coupled with Lys at acetylated pHLIP peptide via self-immolating linker.

DETAILED DESCRIPTION

- (20) The invention provides compounds, peptides with increased affinity to membrane lipid

bilayers at low pH, as well as peptide insertion into and passage across membrane lipid bilayers to facilitate delivery of immuno-stimulatory compounds that promote innate immune responses. Current drugs are not cell permeable, but may enter by transporters or endocytosis.

(21) The invention is used to target tumors with pHLIP® to specifically deliver STING agonists into cancer and immune cells and promote an immune response specifically to the targeted tissue. There are 3 major aspects: i) STING agonists (cyclic di-nucleotides or “CDNs”) are mostly polar molecules, which must be delivered intracellularly to induce the immune reaction; ii) significant activation may be achieved when CDNs are delivered to macrophages within the TME; and ii) CDNs should be targeted to tumors to induce an immune response specifically within tumors only (or predominantly), otherwise the side effects will be devastating. pHLIP® guides or targets CDNs to diseased tissue for intracellular delivery of the CDNs into cancer and immune cells within tumors.

(22) Cyclic dinucleotides are exemplary STING agonists, but other small molecules regulate innate immunity [e.g., DMXAA (also referred to as Vadimezan, ASA-404, CAS Number: 117570-53-3 (in mice) as well as Toll Like Receptor (TLR) agonists, RIG-I]. Each of these classes of immuno-stimulatory compounds train the immune system to attack tumor cells. However, each of these compounds needs to be targeted to a tumor and delivered into cells—a task that is achieved by using pHLIP® technology.

(23) Immuno-Oncology

(24) Immuno-oncology is an emerging field of cancer therapy that aims to activate the immune system specifically in the tumor microenvironment to induce and promote anti-tumor immune responses. Innate immunity is a critical component of host defense, and its function is based on the recognition of pathogen-associated molecular patterns (PAMPs) or danger-associated molecular patterns (DAMPs) through a set of pattern recognition receptors that stimulate the downstream signaling cascades leading to production of pro-inflammatory mediators and type I interferons (IFNs) (Takeuchi O, Akira S, 2010, Pattern recognition receptors and inflammation. Cell 140:805-820). Among intracellular immuno-stimulatory compounds are cytosolic DNA and cyclic dinucleotides recognized by Interferon Regulatory Factor (IRF) or STimulator of INterferon Genes protein (STING). STING (also known as MITA, MPYS, ERIS and TMEM173) is a protein localized predominantly on the endoplasmic reticulum membrane (Ishikawa H, Barber GN, 2008, STING is an endoplasmic reticulum adaptor that facilitates innate immuno signaling. Nature 455:674-678). The ability of STING to induce the production of type I IFNs is used to promote an immune response (Woo SR, Fuertes MB, Corrales L et al, 2014, STING-dependent cytosolic DNA sensing mediates innate immuno recognition of immunogenic tumors. Immunity 41:830-842). STING agonists are cyclic dinucleotides (CDNs) such as cyclic di-GMP (guanosine 5'-monophosphate) (CDG), cyclic di-AMP (adenosine 5'-monophosphate) (CDA), and cyclic GMP-AMP (cGAMP).

(25) To activate the STING pathway and to promote and to induce the immune system for treatment of tumors, STING agonists are delivered to tumors and translocated across membrane into cancer and immune cells using the compositions and methods described herein.

(26) STING in a Tumor Microenvironment

(27) Direct activation of STING in the tumor microenvironment leads to potent and systemic tumor regression and immunity (Corrales et al., Cell Rep. 2015 May 19; 11(7): 1018-1030). Tumor-initiated T cell priming is dependent on IFN- β production by tumor-resident dendritic cells. IFN- β expression is dependent upon activation of the host STING pathway. The highest expression of IFN within tumors upon stimulation with STING agonist was observed in tumor-associated macrophages (TAMs). Four different cell populations from pre-established B16 tumors were sorted: DCs (CD45+ CD11c+ MHCII+), macrophages (CD45+ CD11b+ F4/80+ MHC-II+), T cells (CD45+ CD3+), and endothelial cells (CD45- CD31+). Sorted cells were then stimulated ex vivo with ML RR-S2 CDA (also referred to asDithio-(Rp, Rp)-2',3'-CDA sodium salt; CAS Number:

1638241-89-0) or DMXAA. All subsets expressed IFN- β upon stimulation with the STING agonists. Expression in macrophages was highest, followed by DCs, which were both higher as compared with lymphocytes and endothelial cells.

(28) Synthetic cyclic dinucleotide (CDN) derivatives activate STING alleles, and intratumoral injections of STING agonists such as CDNs are useful as cancer therapeutics; however, their use is hampered by their poor intracellular penetration by passive diffusion across a cell's plasma membranes. Delivery of STING agonists such as CDNs to cells by pHLIP®, e.g., in the form of a composition comprising both a STING agonist and a pHLIP®, significantly increases the introduction/delivery of the STING agonist across the plasma membrane and into the cytosol of a TAM.

(29) pHLIP® Delivery of STING Agonists

(30) The invention provides compositions and methods to target tumors with pHLIP® to specifically deliver STING agonists into cancer and immune cells and promote an immune response specifically to the target tissue. As described above, STING agonists (e.g., cyclic dinucleotides, “CDNs”) are mostly polar molecules, which must be delivered intracellularly to induce the immune reaction. CDNs should be targeted to tumors to induce immune response specifically within tumors only (or predominantly) to avoid or minimize adverse side effects. Intra-tumoral direct injections of CDN can be used. However with pHLIP®, systemic administration is possible—a significant advantage over previous methods.

(31) The methods described herein using pHLIP®-CDN constructs lead to targeted intracellular delivery of CDN into cancer and immune cells within tumors. Systemic or intratumoral injection of pHLIP®-CDN can be used.

(32) pHLIP® Constructs

(33) General representations of pHLIP® compounds comprising pHLIP® peptide and an immuno stimulator cargo molecule are shown in FIGS. 1-16 and described below.

(34) FIG. 1 shows an immuno stimulator cargo (ISC) molecule linked to a pHLIP® peptide:

(35) The combinations shown in FIGS. 2-12 are variations of the scheme shown in FIG. 1.

Exemplary constructs include cGAMP cyclic compounds conjugated to pHLIP®.

(36) One or more modulator molecules (M) are optionally attached to the pHLIP® peptide membrane-inserting end to promote better translocation of an immuno stimulator cargo molecule (FIG. 2). A modulator molecule(s) can be a polar molecule to enhance intracellular delivery of hydrophobic and moderately hydrophobic immuno stimulator cargo molecules. If the cargo is polar (Log P<-0.4), the hydrophobic modulator will increase the Log P of [cargo-modulator] (Log P>-0.4). Alternatively, a modulator molecule(s) can be a non-polar molecule to enhance intracellular delivery of polar immuno stimulator cargo molecules. If the cargo is hydrophobic Log P>2.5, the polar modulator will decrease Log P of [cargo-modulator] (Log P<2.5). Non-limiting examples of modulators are fatty acids, PEG polymers, hydrophobic fluorescent dyes, cyclic peptides. Modulators can alter (increase or decrease) the polarity of the construct/composition by 1%, 5%, 10%, 25%, 50% 75%, 2-fold, 3-fold, 5-fold, 10 fold or more compared to the construct/composition that lacks a modifier.

(37) FIG. 3 shows multiple immuno stimulator cargo molecules linked to a single pHLIP® peptide.

(38) FIG. 4 shows multiple immuno stimulator cargo molecules linked to a single pHLIP® peptide with one or more modulator molecules.

(39) FIGS. 5-8 depict pHLIP® compounds that can carry one or more imaging agents (I) (or other molecules) at pHLIP® peptide membrane-non-inserting end.

(40) FIG. 9 shows two or more pHLIP® peptides with an immuno stimulator cargo linked together via linker molecule

(41) Exemplary constructs include a Var3 pHLIP® sequence

ADDQNPWRAYLDLLFPTDTLLLDLLWCA (SEQ ID NO: 10),

ADQDNPWRAYLDLLFPTDTLLLDLLWCA (SEQ ID NO: 212), or variations thereof, e.g.,

sequences provided in the tables below and in references cited herein (and incorporated by reference).

(42) In preferred embodiments, the cargo is an immune-stimulatory molecule (see, e.g., FIG. 14A-14C). An immuno-stimulatory cargo molecule can target Toll-like Receptors (TLRs), RIG-I-like Receptors (RLRs), or Stimulator of Interferon Genes (STING) or other regulators of the immune response. For example, an immuno-stimulatory cargo molecule binds STING, e.g., an immuno-stimulatory cargo molecule is a STING agonist, e.g., an immuno-stimulatory cargo molecule is cyclic dinucleotide. Non-limiting examples include cyclic di-GMP, cyclic di-AMP and cyclic GMP-AMP, and their derivatives. Exemplary cyclic GMP compound include c[8-AET-G(2',5')pA(3',5')p] (Biolog Cat. No. C 175), c[3'-AHC-G(2',5')pA(3',5')p] (Biolog Cat. No. C 191), and c[G(2',5')p-2'-AHC-A(3',5')p] (Biolog Cat. No. C 192), each of which is commercially available from BIOLOG Life Science Institute, Bremen, Germany. Other examples include their parent compounds: c[G(2',5')pA(3',5')p], c[G(2',5')pA(3',5')p], and c[G(2',5')pA(3',5')p], respectively.

(43) The cargo(s) is linked to pHLIP® peptide(s) via cleavable link(s). For example, the cleavable link can be a disulfide bond, or acid-labile link. In other examples, the cleavable link is a self-immolating link, which allows to release cargo molecule in its non-modified form.

(44) STING Agonists

(45) Immunotherapy includes a number of strategies that harness the immune system to help treat disease. Immunotherapy for cancer relies on the activation of specific immune system cells, e.g., T cells. Cancer drugs called immune checkpoint inhibitors act by removing the brakes imposed on T cells by tumors or by the body's natural mechanisms for limiting their activation to prevent autoimmune disease.

(46) A key player in the body's innate immune response is the STING pathway. The STING pathway was first discovered as a response to viral infections; it senses viral DNA in the cytosol of infected cells. Activation of the protein interferon, a very potent immune system stimulator, is a hallmark of this pathway, which also plays a role in cancer treatment. STING ligands are described in the art, e.g., in U.S. Pat. Nos. 10,045,961; 10,011,630; 9,834,545; 9,642,830; 9,415,045 (each of which is hereby incorporated by reference.) For example, U.S. Pat. No. 10,045,961 describes exemplary STING agonists that include flavonoids, wherein suitable flavonoids include, but are not limited to, 10-(carboxymethyl)-9(10H)acridone (CMA), 5,6-Dimethylxanthenone-4-acetic acid (DMXAA), methoxyvone, 6, 4'-dimethoxyflavone, 4'-methoxyflavone, 3', 6'-dihydroxyflavone, 7, 2'-dihydroxyflavone, daidzein, formononetin, retusin 7-methyl ether, xanthone, or any combination thereof, which are useful in the compositions and methods described herein.

(47) STING acts as a sensor of cytosolic DNA. Bacteria and virus or self-derived DNA in the cytosol activates the STING pathway and promotes the production of type I interferons (IFN-alpha and IFN-beta). STING also participates in cell death signaling through its association with MHC-II and the ERK pathway. STING interacts with DExD/H-Box Helicase 58 (DDX58) DDX58/RIG-I, mitochondrial antiviral-signaling protein (MAVS), signal sequence receptor unit 2 (SSR2), ring finger protein 5 (RNFS), Tripartite Motif Containing 56 (TRIM56), TANK-binding kinase 1 (TBK1), Interferon Induced Protein With Tetratricopeptide Repeats 1 (IFIT1), and Interferon Induced Protein With Tetratricopeptide Repeats 2 (IFIT2). It localizes to the cytoplasm and membranes of the cell, endoplasmic reticulum (ER), and mitochondria; however, in response to DNA stimulation, it translocates to the perinuclear region and interacts with TBK1 kinase. STING's phosphorylation by TBK1 at Ser-358 results in STING activation. STING executes its role by sensing and binding cyclic di-GMP/c-di-GMP and cyclic GMP-AMP/cGAMP. This binding results in the activation of nuclear factor kappa-light chain enhancer of activated B cells (NF-κβ) and interferon regulator factor 3 (IRF3) transcriptional signaling pathways leading to the induction of Type I interferon response.

(48) The U.S. Food and Drug Administration (FDA) has approved several immune checkpoint

drugs for the treatment of various cancers. These drugs target proteins involved in activating the T cell response: programmed death protein 1 (PD-1), programmed death protein ligand 1 (PD-L1), and Cytotoxic T-Lymphocyte Associated Protein 4 (CTLA4). Many clinical trials are testing drugs that target other immune checkpoint proteins (tumor necrosis factor receptor superfamily, member 4 (also referred to as TNFRSF4 or OX40), CD276 (Cluster of Differentiation 276 (also referred to as CD276 or B7-H3), and (lymphocyte activation gene 3 (LAG3), to name just a few), but no notable successes have been reported so far.

(49) T cells are major players in the adaptive immune system—an arm of the immune system that “adapts” or educates itself to recognize highly specific targets (like mutant proteins in cancer cells or specific molecules found on viruses or other pathogens). This adaptive response takes time to mount. In contrast, a more primitive arm of the immune system, the innate immune response, is not very specific, but provides very fast recognition of many kinds of pathogens, including viruses, bacteria, and parasites.

(50) The innate response also plays an integral role in the development of the adaptive response. The innate response involves cells (macrophages and dendritic cells) that may present specific “foreign” molecules to T cells and initiate a specific, adaptive, response. In patients with cancer, the problem is that the adaptive immune response is not activated or inadequately activated.

(51) In the STING pathway, a specific protein called cyclic GMP-AMP Synthase (cGAS) recognizes DNA in the cytosol of an infected cell. cGAS produces a molecule called cGAMP, which activates the protein STING, for which the entire pathway is named STING activates dendritic cells or macrophages, which in turn leads to the activation of the adaptive immune response involving T cells. The STING pathway is also involved in the efficacy of radiation as cancer treatment; the nuclei of cancer cells may break down during radiation, releasing DNA and activating the STING pathway.

(52) Drugs that activate STING or STING agonists include those with a ring-shaped molecular structure known, e.g., cyclic dinucleotide such as cGAMP or engineered agonist molecules that have been shown to activate production of interferon.

(53) Systemic activation of interferon may cause strong inflammatory and autoimmune responses. A significant advantage of the constructs/compositions described herein is that they mediate targeted delivery to tumor cells and/or immune cells within a tumor mass, thereby avoiding or minimizing severe systemic effects. Exemplary STING agonists include ADU-S100 (also referred to as: MIW815, NVP-MIW815, ADUS100) and MK-1454 (Merk—clinical trials identifier NCT03010176).

(54) ##STR00002##

(55) As described above, STING is a central mediator of innate immunity. When stimulated, STING induces the expression of type I interferon, cytokines and T cell recruitment factors that result in the activation of macrophages and dendritic cells, innate effector cells such as natural killer (NK) cells, and priming of tumor-specific T cells. Cyclic dinucleotides (CDNs) and the xanthenone derivative DMXAA bind to and activate STING (Stimulator of interferon genes), leading to a potent type I IFN response.

(56) Thus, CDNs are potent inducers of the innate immune response and are useful as vaccine adjuvants. A synthetic cyclic dinucleotide (CDN) and agonist of STING, has immunoactivating and antineoplastic activities. Upon intratumoral (IT) administration, STING agonist MK-1454 binds to STING and activates the STING pathway, which promotes IKK-related kinase TANK-binding kinase 1 (TBK1) signaling and activates nuclear factor-kappa B (NF-kB) and interferon regulatory factor 3 (IRF3) in immune cells in the tumor microenvironment, thereby leading to the production of pro-inflammatory cytokines, including interferons and a CTL-mediated immune response against tumor cells and tumor cell lysis.

(57) Toll-Like Receptors

(58) Proteins known as toll-like receptors (TLRs), of which there are 10 different types, are found

on the surface and inside of immune cells such as macrophages. TLRs are located on the plasma membrane of cells with the exception of TLR3, TLR7, TLR9 which are localized in the endosomal compartment. Macrophages are immune system cells that not only destroy incoming pathogens, but also may alert the adaptive immune system to an infection. For example, TLR9 recognizes a motif in DNA sequences known as CpG, which is found far more frequently in bacterial than in our own DNA.

(59) When a TLR9 agonist such as a DNA molecule enriched in CpG, is injected into a tumor, rapid activation of macrophages and dendritic cells (DCs) that express TLR9 occurs. The macrophages may directly destroy the infected cells, but, together with DCs, they alert T cells residing in the tumor or draining lymph nodes, and thus promote an adaptive immune response.

(60) Exemplary TLR9 agonists include SD-101, IMO-2125, MGN1703 (Lefitolimod), and DV281. Other TLR agonists, e.g., activators of TLR7/8, include NKTR-262 and MEDI9197. Use of pHLIP® peptides in the compositions or constructs described herein facilitate delivery of TLR agonists to tumors.

(61) pHLIP® Peptides

(62) An example of a wild type (WT) is AEQNPIYWARYADWLFTTPLLALLDLALLVDADEGT (SEQ ID NO: 3) in which AEQNPIY (SEQ ID NO: 4) represents a flanking sequence, WARYADWLFTTPLLALLDLALLV (SEQ ID NO: 5) represents a membrane-inserting sequence, and DAEDEGT (SEQ ID NO: 6) represents a flanking sequence. Other exemplary pHLIP® peptides are shown in the Tables below.

(63) TABLE-US-00001

TABLE	1	Name	Sequence	SEQ ID NO:	Var3-1a
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ADQDNPWRAYLDLLFPTDTLLLDLLWCA	SEQ ID NO: 212	Var3-1b
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ADQDNPWRAYLDLLFPTDTLLLDLLWKA	SEQ ID NO: 213	WT-1
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GGEQNPIYWARYADWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 7	WT-2
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AEQNPIYWARYADWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 8	WT-Cys1
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AAEQNPIYWARYADWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 9	WT-Cys2 Ac-
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AEQNPIYWARYADWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 211	WT-Cys3
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GGEQNPIYWARYADWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 11	Cys-WT1
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Ac-ACEQNPIYWARYADWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 12	Var0-NT
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ACEQNPIYWARYADWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 13	Lys-WT1
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AKEQNPIYWARYADWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 14	Lys-WT2 Ac-
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AKEQNPIYWARYADWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 15	WT-KC
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AAEQNPIYWARYADWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 16	K-WT-C
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AKEQNPIYWARYADWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 17	N-pHLIP®
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ACEQNPIYWARYANWLFTTPLLALLNLALLVDADEGT	SEQ ID NO: 18	N-pHLIP®-b
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ACEQNPIYWARYANWLFTTPLLALLNLALLVDADEGT	SEQ ID NO: 19	K-pHLIP®
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ACEQNPIYWARYAKWLFTTPLLALLKLALLVDADEGT	SEQ ID NO: 20	NNQ
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GGEQNPIYWARYADWLFTTPLLALLDLALLVNANQGT	SEQ ID NO: 21	D25A
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AAEQNPIYWARYADWLFTTPLLALLALALLVDADEGT	SEQ ID NO: 22	D25A-KC Ac-
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AAEQNPIYWARYADWLFTTPLLALLELALLVDADEGT	SEQ ID NO: 23	D14A
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AAEQNPIYWARYAAWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 24	P20A
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AAEQNPIYWARYADWLFTTALLLLDLALLVDADEGT	SEQ ID NO: 25	D25E
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AAEQNPIYWARYADWLFTTPLLALLELALLVDADEGT	SEQ ID NO: 26	D14E
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AAEQNPIYWARYAEWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 27	3D
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AAEQNPIYWARYADWLFTDPLLLLDLALLVDADEGT	SEQ ID NO: 28	R11Q
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GEQNPIYWAQYADWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 29	D25Up
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GGEQNPIYWARYADWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 30	D25Down
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GGEQNPIYWARYADWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 31	D14Up
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GGEQNPIYWARYDAWLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 32	D14Down
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GGEQNPIYWARYAWDLFTTPLLALLDLALLVDADEGT	SEQ ID NO: 33	P20G
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AAEQNPYIYWARYADWLFTTPLLALLVDADDEGT SEQ. ID NO. 34 H1-Cys
 DDEDEDNPYIYWARYADWLFTTPLLALLHGALLVDADECT SEQ. ID NO. 35 H1
 DDEDEDNPYIYWARYADWLFTTPLLALLHGALLVDADET SEQ ID NO: 36 H2-Cys
 DDEDEDNPYIYWARYAHWLFTTPLLALLHGALLVDADEGCT SEQ. ID NO. 37 Cys-H2
 CDDDEDNPYIYWARYAHWLFTTPLLALLHGALLVDADET SEQ ID NO: 38 H2
 DDEDEDNPYIYWARYAHWLFTTPLLALLHGALLVDADEGT SEQ ID NO: 39 H2N-Cys
 DDEDEDNPYIYWARYAHWLFTTPLLALLHGALLVNADECT SEQ. ID NO. 40 H2N
 DDEDEDNPYIYWARYAHWLFTTPLLALLHGALLVNADEGT SEQ ID NO: 41 H2N2-Cys
 DDEDEDNPYIYWARYAHWLFTTPLLALLHGALLVNANECT SEQ. ID NO. 42 H2N2
 DDEDEDNPYIYWARYAHWLFTTPLLALLHGALLVNANEGT SEQ ID NO: 43 1a-Trp
 AEQNPIYWARYADFLFTTPLLALLDLALLVDADET SEQ. ID NO. 44 1b-Trp
 AEQNPIYFARYADWLFTTPLLALLDLALLVDADEGT SEQ. ID NO. 45 1c-Trp
 AEQNPIYFARYADFLFTTPLLALLDLALLWDADDET SEQ. ID NO. 46 Fast-1 or Var1
 AKEDQNPYWARYADWLFTTPLLALLDLALLVDG SEQ. ID NO. 47 Var1-2D1D
 ACEDQNPYWARYADWLFTTPLLALLDLALLVDG SEQ. ID NO. 48 Fast1-Cys or Var1-
 AEDQNPYWARYADWLFTTPLLALLDLALLVDCG SEQ. ID NO. 49 2D1D-Cys Fast1-E-
 Cys or Var1E AEDQNPYWARYADWLFTTPLLALLLELALLVECG SEQ. ID NO. 50
 Fast1-E-Lys AKEDQNPYWARYADWLFTTPLLALLDLALLVG SEQ ID NO: 51 Fast2
 or Var2 AKEDQNPYWRAYADLFTPLTLLDLLALWDG SEQ. ID NO. 52 Fast2-E-Cys
 or Var2E AEDQNPYWARYADWLFTTPLLALLLELALLVCG SEQ ID NO: 53 Var2-2D1D
 ACEDQNPYWRAYADLFTPLTLLDLLALWDG SEQ. ID NO. 54 Var3-3D
 ACDDQNPWRAYLDLLFPTDTLLLDLLW SEQ. ID NO. 55 Var3-3D-cys
 AKDDQNPWRAYLDLLFPTDTLLLDLLWC SEQ ID NO: 56 Var4-3E
 ACEEQNPWRAYLELLFPTETLLLELLW SEQ ID NO: 57 Var5-3Da
 ACDDQNPWARYLDWLFPDTLLLLDL SEQ ID NO: 58 Var6-3Db
 CDNNNPWRAYLDLLFPTDTLLLDLW SEQ ID NO: 59 Var8-3Eb
 CEEQQPWAQYLELLFPTETLLLEW SEQ ID NO: 60 Var9-3Ec
 CEEQQPWRAYLELLFPTETLLLEW SEQ ID NO: 61 Var15-2N
 CDDDDDDNPYIYWARYANWLFTTPLLALLNGALLVEAEET SEQ ID NO: 62 Var16-2P
 CDDDDDDNPYIYWARYAPWLFTTPLLALLPGALLVEAE E SEQ ID NO: 63
 (64) TABLE-US-00002 TABLE 2 Name Sequence SEQ ID NO: Var14-Rev Ac-
 TEDADVLLALDLLLPTTFLWD SEQ. ID NO. 64 AYRAWYPNQECA-Am Sh
 AEQNPIYWARYADWLFTTPL SEQ. ID NO. 65 Sh-Cys AEQNPIYWARYADWLFTTPL
 SEQ. ID NO. 66 Cys-Sh ACEQNPIYWARYADWLFTTPL SEQ. ID NO. 67 Sh-1Trp
 AEQNPIYFARYADWLFTTPL SEQ. ID NO. 68 Sh-W2 AEQNPIYFARYADLLFPTTLAW
 SEQ ID NO. 69 Sh-W1 AEQNPIYWARYADLLFPTTLAF SEQ ID NO. 70 Sh-2W
 AEQNPIYWARYADLLFPTTLAW SEQ ID NO. 71 Sh-1D
 KEDQNPWARYADLLFPTTLAW SEQ. ID NO. 72 Sh-1Db
 KEDQNPWARYADLLFPTTLW SEQ ID NO. 73 Var12-1D
 ACEDQNPWARYADLLFPTTLAW SEQ. ID NO. 74 Var10-2D
 ACEDQNPWARYADWLFPDTLLLLD SEQ. ID NO. 75 Var13-1E
 ACEEQNPWARYAEELLFPTTLAW SEQ. ID NO. 76 Var11-2E
 ACEEQNPWARYAEWLFPDTLLLLLE SEQ. ID NO. 77 Var7-3E
 ACEEQNPWARYLEWLFPDTETLLLEL SEQ. ID NO. 78 Var7-3Eb
 ACEEQNPQAEYAEWLFPDTLLLLLE SEQ ID NO: 79 * Ac means Acetylated N-terminus;
 and Am means Amidated C-terminus

(65) TABLE-US-00003 TABLE 3 Coded and exemplary non-coded amino acids including L-
 isomers, D- isomers, alpha-isomers, beta-isomers, glycol-, and methyl- modifications. No. Abbrev
 Name 1 Ala Alanine 2 Arg Arginine 3 Asn Asparagine 4 Asp Aspartic acid 5 Cys Cysteine 6 Gln
 Glutamine 7 Glu Glutamic acid 8 Gly Glycine 9 His Histidine 10 Ile Isoleucine 11 Leu Leucine 12

Lys Lysine 13 Met Methionine 14 Phe Phenylalanine 15 Pro Proline 16 Ser Serine 17 Thr Threonine 18 Trp Tryptophan 19 Tyr Tyrosine 20 Val Valine 21 Sec Selenocysteine 22 Sem Selenomethionine 23 Pyl Pyrrolysine 24 Aad Alpha-amino adipic acid 25 Acpa Amino-caprylic acid 26 Aecys Aminoethyl cysteine 27 Afa Aminophenyl acetate 28 Gaba Gamma-aminobutyric acid 29 Aiba Aminoisobutyric acid 30 Aile Alloisoleucine 31 Aig Allylglycine 32 Aba Amino-butyric acid 33 Aphe Amino-phenylalanine 34 Brphe Bromo-phenylalanine 35 Cha Cyclo-hexylalanine 36 Cit Citrulline 37 Clala Chloroalanine 38 Cie Cycloleucine 39 Clphe Fencionine (or chlorophenylalanine 40 Cya Cysteic acid 41 Dab Diaminobutyric acid 42 Dap Diaminopropionic acid 43 Dap Diaminopimelic acid 44 Dhp Dehydro-proline 45 Dhphe DOPA (or 3,4-dihydroxyphenylalanine) 46 Fphe Fluorophenylalanine 47 Gaa Glucosaminic acid 48 Gla Gamma-carboxyglutamic acid 49 Hag Homoarginine 50 Hlys Hydroxylysine 51 Hnvl Hydroxynorvaline 52 Hog Homoglutamine 53 Hoph Homophenylalanine 54 Has Homoserine 55 Hse Homocysteine 56 Hpr Hydroxyproline 57 Iphe Iodo-phenylalanine 58 Ise Isoserine 59 Mle Methyl-leucine 60 Msmet Methionine-methylsulfonium chloride 61 Nala Naphthyl-alanine 62 Nle Norleucine (or 2-amino hexanoic acid) 63 Nmala N-methyl-alanine 64 Nva Norvaline (or 2-aminopentanoic acid) 65 Obser O-benzyl-serine 66 Obtyr O-benzyl-tyrosine 67 Oetyr O-ethyl-tyrosine 68 Omser O-methyl-serine 69 Omthr O-methy-threonine 70 Omtyr O-methyl-tyrosine 71 Orn Ornithine 72 Pen Penicillamine 73 Pga Pyroglutamic acid 74 Pip Pipecolic acid 75 Sar Sarcosine 76 Tfa Trifluoro-alanine 77 Thphe Hydroxy-Dopa 78 Vig Vinylglycine 79 Aaspa Amino-aminoethylsulfanylpropanoic acid 80 Ahdna Amino-hydroxy-dioxanonanolic acid 81 Ahoha Amino-hydroxy-oxahexanoic acid 82 Ahsopa Amino-hydroxyethylsulfanylpropanoic acid 83 Tyr(Me) Methoxyphenyl-methylpropanyl oxycarbonylamino 84 MTrp propanoic acid 85 pTyr Methyl-tryptophan 86 pSer Phosphorylated Tyr 87 pThr Phosphorylated Ser 88 BLys Phosphorylated Thr 89 Hyp BiotinLys 90 Phg Hydroproline 91 Cha Phenylglycine 92 Chg Cyclohexyl-alanine 93 Nal Cyclohexylglycine 94 Pal Naphthylalanine 95 Pra Pyridyl-alanine 96 Gly(allyl) Propargylglycine 97 Pen Pentenoic acid 98 MetO Penicillamine 99 Pca Methionine sulfoxide 100 Ac-Lys Pyroglutamic acid Acetylation of Lys

(66) TABLE-US-00004 TABLE 4 Non-limiting examples of protonatable residues and their substitutions including L-isomers, D-isomers, alpha-isomers, and beta-isomers. Original Residue Exemplary amino acids substitution Asp (D) Glu (E); Gla (Gla); Aad (Aad) Glu (E) Asp (D); Gla (Gla); Aad (Aad)

(67) TABLE-US-00005 TABLE 5 Examples of coded amino acid substitutions Original Residue Substitution Ala (A) Gly; Ile; Leu; Met; Phe; Pro; Trp; Tyr; Val Arg (R) Lys Asn (N) Gln; His Asp (D) Glu Cys (C) Ser; Met Gln (Q) Asn; His Glu (E) Asp Gly (G) Ala; Ile; Leu; Met; Phe; Pro; Trp; Tyr; Val His (H) Asn; Gln Ile (I) Ala; Gly; Leu; Met; Phe; Pro; Trp; Tyr; Val Leu (L) Ala; Gly; Ile; Met; Phe; Pro; Trp; Tyr; Val Lys (K) Arg Met (M) Ala; Gly; Leu; Ile; Phe; Pro; Trp; Tyr; Val Phe (F) Ala; Gly; Leu; Ile; Met; Pro; Trp; Tyr; Val Pro (P) Ala; Gly; Leu; Ile; Met; Phe; Trp; Tyr; Val Ser (S) Thr Thr (T) Ser Trp (W) Ala; Gly; Leu; Ile; Met; Pro; Phe; Tyr; Val Tyr (Y) Ala; Gly; Leu; Ile; Met; Pro; Phe; Trp; Val Val (V) Ala; Gly; Leu; Ile; Met; Pro; Phe; Trp; Tyr

(68) TABLE-US-00006 TABLE 6 Non-limiting examples of membrane-inserting sequences belonging to different groups of pHLIP® peptides. Each protonatable residue (shown in bold) could be replaced by its substitution from Table 4. Each non-polar residue could be replaced by its coded amino acid substitution from Table 5, and/or non-coded amino acid substitutions from Table 3. Groups Sequences (SEQ ID NO:) WT-BRC WARYADWLFTTPLL**LL**DLALL (SEQ ID NO: 80) YARYADWLFTTPLL**LL**DLALL (SEQ ID NO: 81) WARYSDWLFTTPLL**LY**DLGLL (SEQ ID NO: 82) WARYTDWFTTPLL**LY**DLALLA (SEQ ID NO: 83) WARYTDWLFTTPLL**LY**DLGLL (SEQ ID NO: 84) WARYADWLFTTPLL**LL**DL**S**LL (SEQ ID NO: 85) WT-BRC LLAL**D**LL**LL**PTTFLW**D**AYRAW (SEQ ID NO: 86) Reverse

LLALDLLLPTTFLWDAYRAW (SEQ ID NO: 87) LLGLDYLLLPPTTFLWDSYRAW
 (SEQ ID NO: 88) ALLALDYLLLPPTTFLWDYRAW (SEQ ID NO: 89)
 LLGLDYLLLPPTTFLWDYRAW (SEQ ID NO: 90) LLSLDLLLPPTTFLWDAYRAW
 (SEQ ID NO: 91) ATRAM GLAGLLGLEGLLGLPLGLLEGLWLGL (SEQ ID NO:
 92) ATRAM LGLWLGEELLGLPLGLLGELGLLGALG (SEQ ID NO: 93) Reverse Var3
 WRAYLDLLFPDTLLLDLLW (SEQ ID NO: 94) Var3
 WLLDLLLTDTPFLLDLYARW (SEQ ID NO: 95) Reverse Var7
 WARYLEWLFPTETLLEL (SEQ ID NO: 96) WAQYLELLFPPTETLLEW (SEQ
 ID NO: 97) Var7 LELLLTETPFLELYRAW (SEQ ID NO: 98) Reverse
 WELLTETPFLELYQAW (SEQ ID NO: 99) Single WLFTTPLLNGALLVE
 (SEQ ID NO: 100) D/E WLFTTPLLPGALLVE (SEQ ID NO: 101)
 WARYADLLFPPTLAW (SEQ ID NO: 102) Single EVLLAGNLLLPPTFLW (SEQ
 ID NO: 103) D/E EVLLAGPLLPTTFLW (SEQ ID NO: 104) Reverse
 WALTPFLLDAYRAW (SEQ ID NO: 105) pHLP[®]-
 NLEGFFATLGGEIALWSLVVLAIE (SEQ ID NO: 106) Rho
 EGFPATLGGEIALWSDVLAIE (SEQ ID NO: 107) EGFPATLGGEIPLWSDVLAIE
 (SEQ ID NO: 108) pHLP[®]- EIALVVLWSLAIEGGLTAFFGELN (SEQ ID NO:
 109) Rho EIALVVDSWLAIEGGLTAFFGE (SEQ ID NO: 110) Reverse
 EIALVVDSWLPIEGGLTAFFGE (SEQ ID NO: 111) pHLP[®]-
 ILDLVFGLLFAVTSVDFLVQW (SEQ ID NO: 112) CA9 pHLP[®]-
 WQVLF[®]DVSTVAFLGFLVLDLI (SEQ ID NO: 113) CA9 Reverse

(69) TABLE-US-00007 TABLE 7 Non-limiting examples of pHLP[®] sequences.
 A cysteine, a lysine, an azido-modified amino acid, or an alkynyl modified
 amino acid can be incorporated at the N-terminal (first 6 residues) or C-
 terminal (last 6 residues) parts of the peptides for conjugation with a cargo,
 and a linker. SEQ ID NO: Name Sequence SEQ ID NO:

AEQNPIYWARYADWLFTTPLL [®] LDLALLVDADET	SEQ ID NO: 114	WT-2D
AEQNPIYWARYAEWLFTTPLL [®] LELALLVEAEET	SEQ ID NO: 115	WT-6E
ADDQNPWRAYLDLLFPDTTDL [®] LDL [®] LDLWDADDET	SEQ ID NO: 116	WT-3D
ADDQNPWRAYLDLLFPDTTDL [®] LDL [®] LDLWDADDET	SEQ ID NO: 117	WT-9E
AEEQNPWRAYLELLFPETTEL [®] LELLWEAEET	SEQ ID NO: 118	WT-GlaD
AEQNPIYWARYAGlaWLFTTPLL [®] LDLALLVDADET	SEQ ID NO: 119	WT-DGla
AEQNPIYWARYADWLFTTPLL [®] LGlaLALLVDADET	SEQ ID NO: 120	WT-2Gla
AEQNPIYWARYAGlaWLFTTPLL [®] LGlaLALLVDADET	SEQ ID NO: 121	WT-AadD
AEQNPIYWARYAAadWLFTTPLL [®] LDLALLVDADET	SEQ ID NO: 122	WT-DAad
AEQNPIYWARYADWLFTTPLL [®] LAadLALLVDADET	SEQ ID NO: 123	WT-2Aad
AEQNPIYWARYAAadWLFTTPLL [®] LAadLALLVDADET	SEQ ID NO: 124	WT-GlaAad
AEQNPIYWARYAGlaWLFTTPLL [®] LAadLALLVDADET	SEQ ID NO: 125	WT-AadGla
AEQNPIYWARYAAadWLFTTPLL [®] LGlaLALLVDADET	SEQ ID NO: 126	WT-1
GGEQNPIYWARYADWLFTTPLL [®] LDLALLVDADEGT	SEQ ID NO: 127	WT-2
GGEQNPIYWARYADWLFTTPLL [®] LDLALLVDADEGT	SEQ ID NO: 128	WT-3
AAEQNPIYWARYADWLFTTPLL [®] LDLALLVDADEGT	SEQ ID NO: 129	WT-4
AEQNPIYWARYADWLFTTPLL [®] LDLALLVDADEGT	SEQ ID NO: 130	WT-2N
AEQNPIYWARYANWLFTTPLL [®] NLALLVDADEGT	SEQ ID NO: 131	WT-2K
AEQNPIYWARYAKWLFTTPLL [®] KLALLVDADEGT	SEQ ID NO: 132	WT-2DNANQ
GGEQNPIYWARYADWLFTTPLL [®] LDLALLVNANQGT	SEQ ID NO: 133	WT-D25A
AAEQNPIYWARYADWLFTTPLL [®] LALALLVDADEGT	SEQ ID NO: 134	WT-D14A
AAEQNPIYWARYAAWLFTTPLL [®] LDLALLVDADEGT	SEQ ID NO: 135	WT-P20A
AAEQNPIYWARYADWLFTTALL [®] LDLALLVDADEGT	SEQ ID NO: 136	WT-D25E
AAEQNPIYWARYADWLFTTPLL [®] LELALLVDADEGT	SEQ ID NO: 137	WT-D14E
AAEQNPIYWARYAEWLFTTPLL [®] LDLALLVDADEGT	SEQ ID NO: 138	WT-3D-2

AAEQNPYARYADWLFTTDLALLVDADEGT SEQ ID NO: 139 WT-R11Q
GEQNPIYWAQYADWLFTTPLLLLDLALLVDADEG SEQ ID NO: 140 WT-D25Up
GGEQNPIYWARYADWLFTTPLLLLDLALLVDADEG SEQ ID NO: 141 WT-D25Down
GGEQNPIYWARYADWLFTTPLLLLDLALLVDADEG SEQ ID NO: 142 WT-D14Up
GGEQNPIYWARYDAWLFTTPLLLLDLALLVDADEGT SEQ ID NO: 143 WT-D14Down
GGEQNPIYWARYAWDLFTTPLLLLDLALLVDADEG SEQ ID NO: 144 WT-P20G
AAEQNPYARYADWLFTTGLLLLDLALLVDADEGT SEQ ID NO: 145 WT-DH
DDDEDNPIYWARYADWLFTTPLLLLLHGALLVDAD SEQ ID NO: 146 WT-2H
DDDEDNPIYWARYAHWLFTTPLLLLLHGALLVDADE SEQ ID NO: 147 WT-L16H
CEQNPIYWARYADWHFTTPLLLLDLALLVDADE SEQ ID NO: 148 WT-1Wa
AEQNPIYWARYADFLFTTPLLLLDLALLVDADET SEQ ID NO: 149 WT-1Wb
AEQNPIYFARYADWLFTTPLLLLDLALLVDADE SEQ ID NO: 150 WT-1Wc
AEQNPIYFARYADFLFTTPLLLLDLALLWDADDET SEQ ID NO: 151 WT-W6
ADNNPWIYARYADLTTFPLLLLDLALLVDFDD SEQ ID NO: 152 WT-W17
ADNNPFIYARYADLTWPLLLLDLALLVDFDD SEQ ID NO: 153 WT-W30
ADNNPFIYARYADLTTFPLLLLDLALLVDWDD SEQ ID NO: 154 WT-W17-P7
ADNNPFIYARYADLTWILLLLDLALLVDFDD SEQ ID NO: 155 WT-W39-R11
ADNNPFIYAYRADLTTFPLLLLDLALLVDWDD SEQ ID NO: 156 WT-W30-R15
ADNNPFIYATYADLRTFPLLLLDLALLVDWDD SEQ ID NO: 157 WT-Rev Ac-
TEDADVLLALDLLLLPTTFLWDAYRAWYPNQEA-Am SEQ ID NO: 158 Var1-3D
AEDQNPYARYADWLFTTPLLLLDLALLVD SEQ ID NO: 159 Var1-1D2E
AEDQNPYARYADWLFTTPLLLLELALLVE SEQ ID NO: 160 Var2-3D
AEDQNPYWRAYADLFTPLTLLDLLALWD SEQ ID NO: 161 Var3-3D
ADDQNPWRAYLDLLFPTDTLLLDLLW SEQ ID NO: 162 Var3-WT
ADDQNPWRAYLDLLFPTDTLLLDLLWDADE SEQ ID NO: 163 Var3-Gla2D
ADDQNPWRAYLGlaLLFPTDTLLLDLLW SEQ ID NO: 164 Var3-DGlaD
ADDQNPWRAYLDLLFPTGlaTLLLDLLW SEQ ID NO: 165 Var3-2DGla
ADDQNPWRAYLDLLFPTDTLLLGlaLLW SEQ ID NO: 166 Var3-2GlaD
ADDQNPWRAYLGlaLLFPTGlaTLLLDLLW SEQ ID NO: 167 Var3-GlaDGla
ADDQNPWRAYLGlaLLFPTDTLLLGlaLLW SEQ ID NO: 168 Var3-D2Gla
ADDQNPWRAYLDLLFPTGlaTLLLGlaLLW SEQ ID NO: 169 Var3-3Gla
ADDQNPWRAYLGlaLLFPTGlaTLLLGlaLLW SEQ ID NO: 170 Var3-Aad2D
ADDQNPWRAYLAadLLFPTDTLLLDLLW SEQ ID NO: 171 Var3-DAadD
ADDQNPWRAYLDLLFPTAadTLLLDLLW SEQ ID NO: 172 Var3-2DAad
ADDQNPWRAYLDLLFPTDTLLLAadLLW SEQ ID NO: 173 Var3-2AadD
ADDQNPWRAYLAadLLFPTAadTLLLDLLW SEQ ID NO: 174 Var3-AadDAad
ADDQNPWRAYLAadLLFPTDTLLLAadLLW SEQ ID NO: 175 Var3-D2Aad
ADDQNPWRAYLDLLFPTAadTLLLAadLLW SEQ ID NO: 176 Var3-3Aad
ADDQNPWRAYLAadLLFPTAadTLLLAadLLW SEQ ID NO: 177 Var3-GlaAadD
ADDQNPWRAYLGlaLLFPTAadTLLLDLLW SEQ ID NO: 178 Var3-GlaDAad
ADDQNPWRAYLGlaLLFPTDTLLLAadLLW SEQ ID NO: 179 Var3-2GlaAad
ADDQNPWRAYLGlaLLFPTGlaTLLLAadLLW SEQ ID NO: 180 Var3-AadGlaD
ADDQNPWRAYLAadLLFPTGlaTLLLDLLW SEQ ID NO: 181 Var3-AadDGla
ADDQNPWRAYLAadLLFPTDTLLLGlaLLW SEQ ID NO: 182 Var3-GlaAadGla
ADDQNPWRAYLGlaLLFPTAadTLLLGlaLLW SEQ ID NO: 183 Var3-GLL
GEEQNPWLGayLDLLFPLELLGLELGLW SEQ ID NO: 184 Var3-M
ADDDDDDPWQAYLDLLFPTDTLLLDLLW SEQ ID NO: 185 Var4-3E
AEEQNPWRAYLELLFPTETLLLELLW SEQ ID NO: 186 Var5-3Da
ADDQNPWARYLDWLFTDTLLLDL SEQ ID NO: 187 Var6-3Db
DNNNPWRAYLDLLFPTDTLLLDW SEQ ID NO: 188 Var7-3E

AEEQNPWARYLEWLFPTTLLLEL SEQ ID NO: 189 Var7-M
 DDDDDDDPWQAYLDLFTDTLALDLW SEQ ID NO: 190 Var8-3E
 EEQQPWAQYLELLFPTTLLLEW SEQ ID NO: 191 Var9-3E
 EEQQPWRAYLELLFPTTLLLEW SEQ ID NO: 192 Var10-2D
 AEDQNPWARYADWLFPTTLLLLD SEQ ID NO: 193 Var11-2E
 AEEQNPWARYAEWLFPTTLLLEL SEQ ID NO: 194 Var12-1D
 AEDQNPWARYADLLFPTTTLAW SEQ ID NO: 195 Var13-1E
 AEEQNPWARYAELLFPTTTLAW SEQ ID NO: 196 Var15-2N
 DDDDDNPNYWARYANWLFTTPLLNGALLVEAEET SEQ ID NO: 197 Var16-2P
 DDDDDNPNYWARYAPWLFTTPLLPGALLVEAEET SEQ ID NO: 198 Var17
 AEQNPIYFARYADFLFTTPLLDDLALLWDADET SEQ ID NO: 199 Var18
 AEQNPIYWARYADFLFTTPLLDDLALLVDADET SEQ ID NO: 200 Var19a
 AEQNPIYWARYADWLFTTPL SEQ ID NO: 201 Var20 AEQNPIYFARYADLLFPTTTLAW
 SEQ ID NO: 202 Var21 AEQNPIYWARYADLLFPTTTLAF SEQ ID NO: 203 Var22
 AEQNPIYWARYADLLFPTTTLAW SEQ ID NO: 204 Var23 AEQNPIYFARYADWLFTTPL
 SEQ ID NO: 205 Var24 EDQNPWARYADLLFPTTTLAW SEQ ID NO: 206 ATRAM
 GLAGLAGLLGLEGLLGLPLGLEGLWLGLELEGN SEQ ID NO: 207 pHLIP®-CA9
 EQNPIYILDVEGLLFAVTSVDELVQWDDAGD SEQ ID NO: 208 pHLIP®-Rho
 NLEGFFATLGGEIALWSLVVLAIE SEQ ID NO: 209 pHLIP®-RhoM1
 NNEGFFATLGGEIALWSDVVLAIE SEQ ID NO: 210 pHLIP®-RhoM2
 DNNEGFFATLGGEIPLWSDVVLAIE

Example 1: pHLIP® Peptides

(70) pHLIP® peptides are described here and in U.S. Pat. Nos. 9,814,781 and 9,289,508 (hereby incorporated by reference) as well as U.S. Patent Publication 20180117183, 20180064648, 20180221500, 20180117183, 20180064648, 20160256560, 20150191508, 20150051153, and 20120142042, 20120039990, and 20080233107, each of which is hereby incorporated by reference.

(71) STING is known in the literature but the use of STING agonists has been hampered by cell delivery issues. FIGS. 13-16 show exemplary pHLIP®-linker-Cargo constructs, e.g., with the cargo being an immune-stimulatory compound. As described above, it is difficult to achieve delivery of charged molecules to go through a cell membrane. Constructs described here, e.g., a pHLIP®-linker-Cargo construct, mediate cancer and immune cell targeting (due to their surface low pH)) and also avoid or minimize a global auto-immune response with a STING agonist (for instance cGAMP). The problem of targeting the cargo, e.g., an immune-stimulating drug, and getting the cargo inside the cell is solved by the pHLIP®/drug compositions and methods described herein.

(72) A linker could be relatively small, e.g., only a few atoms, to a rather large polymer of 4-5 kDa. FIGS. 13-16 show an exemplary heterobifunctional linker that reacts on one end with a free thiol to spontaneously form a disulfide bond, with thiopyridine as a leaving group, and on the other end reacts with activated with amine or hydroxyl groups in the presence of DIPEA, and in some cases DMAP or other activator base, to form a carbamate or carbonate, respectively. This material can be used if pHLIP® (A) is protected at its amino terminus, such as with N-acetylation. This material can also be reacted with pHLIP® bearing a cysteine residue or with a thiol-bearing linker for subsequent conjugation to pHLIP®, and forms a conjugate by disulfide exchange with thiopyridine as a leaving group. This material can be used to form a conjugate with pHLIP® bearing a lysine residue, if pHLIP® is protected at its amino terminus, such as with N-acetylation.

(73) In some examples, a succinimidyl 3-(2-pyridyldithio)propionate (SPDP) cross-linker is used. SPDP is a short-chain crosslinker for amine-to-sulphydryl conjugation via NHS-ester and pyridyldithiol reactive groups that form cleavable (reducible) disulfide bonds with cysteine sulphydryls. SPDP is used to activate NH₂ derivative of cCDN (of e.g., c[3'-AHC-G(2',5')pA(3',5')p], or c[G(2',5')p-2'-AHC-A(3',5')p]), purify and exchange disulfide with SH of

single Cys at the C-terminus of pHLIP® to obtain pHLIP®-S-S-Ccdn.

(74) In some examples, the following cross-linkers can be used: LC-SPDP (succinimidyl 6-(3(2-pyridyldithio)propionamido)hexanoate); Sulfo-LC-SPDP (sulfosuccinimidyl 6-(3'-(2-pyridyldithio)propionamido)hexanoate); PEG4-SPDP (PEGylated, long-chain SPDP crosslinker); PEG12-SPDP (PEGylated, long-chain SPDP crosslinker); SMCC (succinimidyl 4-(N-maleimidomethyl)cyclohexane-1-carboxylate); Sulfo-SMCC (sulfosuccinimidyl 4-(N-maleimidomethyl)cyclohexane-1-carboxylate); SMPT (4-succinimidyl oxycarbonyl- α -methyl- α -(2-pyridyldithio)toluene); DTME (dithiobismaleimidoethane).

(75) The invention may encompass the following embodiments.

(76) A compound of formula:

Peptide-Mod-Linker-Drug (1),

wherein:

(77) Peptide is a pHLIP® peptide,

(78) Mod is a modulator, and it is optional. It comprises chemical entity to modulate overall polarity of Linker-Drug for optimized intracellular delivery by pHLIP®. To achieve optimized intracellular delivery the overall polarity of Mod-Linker-Drug measured by Log P, where P is the measured octanol-water partition coefficient, is preferably in the range $-1 < \text{Log P} < 1$. If the cargo is polar ($\text{Log P} < -0.4$), the hydrophobic modulator will increase the Log P of [cargo-modulator] ($\text{Log P} > -0.4$). If the cargo is hydrophobic $\text{Log P} > 2.5$, the polar modulator will decrease Log P of [cargo-modulator] ($\text{Log P} < 2.5$).

(79) In some cases, an immune-stimulatory compound or drug is moderately hydrophobic. The average value of Log P for drugs is about 2-3. Exemplary cargo compounds, e.g., immune-stimulatory compounds or drugs, are polar, moderately hydrophobic or hydrophobic as defined by the following characteristics. Polar: $\text{Log P} < -0.4$; Moderately hydrophobic: $2.5 < \text{Log P} < -0.4$; and Hydrophobic: $\text{Log P} > 2.5$. The polarity and/or hydrophobicity of a drug or compound to be delivered is measured using methods known in the art, e.g., by determining Log P, in which P is octanol-water partition coefficient. A substance is dissolved into octanol-water mixture, mixed and allowed to come to equilibration. The amount of substance in each (or one) phases is then measured. The measurements itself could be in a number of ways known in the art, e.g., by measuring absorbance, or determining the amount using NMR, HPLC, or other known methods.

(80) Drug comprises or consists of a drug or compound with anticancer activity.

(81) Linker comprises a covalent bond or a chemical linker such that (1) is selected from the following (for example, where "Drug" includes an Immuno-Stimulatory Compound):

(82) ##STR00003##

(83) each occurrence of y may be present or absent and is independently an integer ranging from 1 to 4;

(84) each occurrence of X is independently selected from the group consisting of CH_{sub.2}, CH(alkyl), and C(alkyl)_{sub.2};

(85) each occurrence of B may be present or absent and is independently selected from the group consisting of alkyl, aryl, and PEG;

(86) bond a is formed between the sulfur and the thiol substituent of a cysteine residue in A;

(87) bond b is formed between the carbon and a substituent on Drug (e.g., an immuno-stimulatory compound), wherein the substituent is selected from the group consisting of hydroxyl, carbonyl, amine, amide, sulfate, sulfonamide, phosphate, and phosphoramidate;

(88) bond c is formed between the carbonyl and a substituent on Drug (e.g., an immuno-stimulatory compound), wherein the substituent is selected from the group consisting of primary amine, secondary amine, and hydroxyl;

(89) bond d is formed between B and an amino acid residue in A, wherein the amino acid is selected from the group consisting of serine, threonine, tyrosine, tryptophan, histidine, lysine, and cysteine and comprises an amide, ester, carbamate, carbonate, or maleimide bond; and

(90) Drug is an anticancer drug with an immune associated mechanism of action, including but not limited to the group consisting of RIG-I agonist, TLR-7 agonist, and STING agonist; or a salt, solvate, enantiomer, diastereoisomer, geometric isomer, or tautomer thereof.

(91) A compound of formula (1), wherein Drug (e.g., an immuno-stimulatory compound) is a cyclic dinucleotide STING agonist.

(92) A compound of formula (1), wherein Drug (e.g., an immuno-stimulatory compound) is selected from the group consisting of cyclic dinucleotides: c-GAMP (or cyclic GMP-AMP); c-diAMP (or cyclic di-AMP); c-diGMP (or cyclic di-GMP):

(93) ##STR00004##

and their analogs and derivatives. The non-limiting examples are cyclic di-nucleotides in which: one or both phosphate moieties, where one of two exocyclic oxygens is replaced by sulfur for improved stability against degradation by phosphodiesterases; the hydrogen in position 8 of one or both guanine nucleobase is modified for conjugation with one or two pHLIP® peptides; the hydrogen in position 8 of one of both adenine nucleobase is modified for conjugation with one or two pHLIP® peptides; the hydrogen in position 8 of both adenine and guanine nucleobases is modified for conjugation with one or two pHLIP® peptides; a spacer with a terminal reactive group is attached to one or both ribose 3'-hydroxy group of the guanosine for conjugation with one or two pHLIP® peptides; a spacer with a terminal reactive group is attached to one or both ribose 2'-hydroxy group of the adenosine for conjugation with one or two pHLIP® peptides; 2'-deoxy analogues; both the N1 and the N6 nitrogen atoms in the adenine nucleobase are connected by an etheno bridge forming a tricyclic ring system; synthetic cGAMP linked via two 2',5' phosphodiester bonds; synthetic c-diAMP containing two distinct phosphodiester linkages similar to the cGAMP; metabolic degradation products of c-GAMP, c-diAMP, c-diGMP.

Example 2: pHLIP®-Mediated Tumor Targeting and Cytoplasmic Delivery of CDNs

(94) pHLIP-CDNs (cyclic di-nucleotides), STING agonists, such as pHLIP-S-S-cGAMP (schematic below), are synthesized and purified by CheminPharma, Inc. STING agonists are coupled to C-terminal, membrane-inserting end of pHLIP® peptide via self-immolative linkers and are released in cytoplasm in their non-modified form to be able effectively activate STING pathway.

(95) ##STR00005##

Chemical structures of pHLIP-S-S-cGAMP conjugates cGAMP is coupled with Cys at pHLIP® peptide (A) or Lys at acetylated pHLIP® peptide (B).

(96) pHLIP-S-S-cGAMP are tested on B16-Blue ISG and B16-Blue ISG-KO-STING cells (Invitrogen). B16-Blue ISO is a murine melanoma cell line stably transfected with a secreted embryonic alkaline phosphatase (SEAP) gene under the control of the interferon-inducible ISG54 promoter enhanced by a multimeric interferon-stimulated response element (ISRE) (<http://www.invivogen.com/b16-blue-isg>). B16-Blue ISO-KO-STING cells are derived from B16-Blue ISO by stable knockout of the stimulator of interferon genes (STING) gene (<http://www.invivogen.com/b16-blue-isg-ko-sting>). B16-Blue ISG cells express the secreted embryonic alkaline phosphatase (SEAP) reporter gene under control of interferon-inducible ISG54 promoter. Stimulation of B16-Blue ISG with CDNs triggers the production of interferons, leading to activation of I-ISG54 promoter and the production of SEAP in the supernatant. Level of SEAP in supernatant can be determined by measuring of absorbance at 655 nm. B16-Blue ISG and B16-Blue ISG-KO-STING cells are seeded 30,000/well in 96-well plate. The next day cells are treated with different concentrations of ADU-S100 (positive control) and pHLIP-S-S-cGAMP in 80 µl DMEM without PBS, pH 6.2 for 2 hour followed by addition of 40 µl of DMEM/20% FBS, pH 7.4 for 22 hours. The 50 µl of supernatant is taken from each well, mixed with 150 µl QUANTI-Blue solution (Invivogen) and incubated for 1.5-2 hours at 37° C., and optical density is measured at 655 nm. When STING pathway is activated in B16-Blue ISG cells, the absorbance readings increase

(97) Self-immolating chemistry, e.g., linker(s), is used to release STING agonists in their non-

modified form.

(98) pHLIP-S-S-cGAMP agonists are given as multiple intraperitoneal (IP) or intratumoral (IT) injections into mice bearing HeLa cervical tumor in flanks of female athymic nude mice. When tumor is reached size of about 1 cm.^{sup.3} (about 1 g) in the control (non-treated) group, the animals are sacrificed; tumors are collected and weighted. About 40-60% of tumor weight reduction is observed after administration of pHLIP-S-S-cGAMP agonist.

(99) pHLIP®-Mediated Intracellular Delivery of Immuno-Stimulatory Compounds

(100) Provided herein is a composition comprising an immuno-stimulatory compound and a pHLIP® peptide. In some examples, the composition has the following structure: Peptide—Linker-ISC, wherein “Peptide” is a pHLIP® peptide comprising the sequence ADQDNPWRAYLDLLFPTDTLLLDLLWCA (SEQ ID NO: 212) or ADDQNPWRAYLDLLFPTDTLLLDLLWCA (SEQ ID NO: 10) or ADQDNPWRAYLDLLFPTDTLLLDLLWKA (SEQ ID NO: 213) or ADDQNPWRAYLDLLFPTDTLLLDLLWKA (SEQ ID NO: 217), wherein “Linker” is a cleavable linker, wherein “ISC” is an immuno-stimulatory compound, and wherein each “—” is a covalent bond.

(101) In some examples, the immuno-stimulatory compound comprises cyclic dinucleotides (CDNs), or derivatives thereof. In other aspects, the immune-stimulatory compound comprises cyclic purine dinucleotide. Also, as describe herein, the immuno-stimulatory compound includes a cyclic purine dinucleotide which binds to stimulator of interferon genes (STING). In still other examples, the immuno-stimulatory compound comprises a cGAMP, 3',5'-cyclic diadenylic acid (c-di-AMP), or a cyclic diguanylate (c-di-GMP) cyclic compound, or a derivative thereof.

(102) Optionally, the linker, as described herein is a cleavable linker. For example, the linker may include a disulfide bond or an acid-labile bond. In other examples, the linker may be self-immolating.

(103) Other exemplary compositions are described below. For example, the composition comprising an immuno-stimulatory compound and a pHLIP® peptide is exemplified by the following structure: Peptide-Linker-ISC. In one option, the composition further comprises a modulator of polarity.

(104) In other examples, the composition described herein includes 2 or more pHLIP® peptides. For example, the composition comprising 2 or more pHLIP® peptides has the following structure: Peptide1-Link-Peptide2. In aspects, the “Peptide1” is a first pHLIP® peptide comprising the sequence ADDQNPWRAYLDLLFPTDTLLLDLLWXA (SEQ ID NO: 1) or ADQDNPWRAYLDLLFPTDTLLLDLLWXA (SEQ ID NO: 214), “Peptide2” is a second pHLIP® peptide comprising the sequence ADDQNPWRAYLDLLFPTDTLLLDLLWXA (SEQ ID NO: 1) or ADQDNPWRAYLDLLFPTDTLLLDLLWXA (SEQ ID NO: 214). For example, “X” indicates any amino acid residue, including include a lysine (Lys), a cysteine (Cys), or an Azido-containing amino acid, “Link” is a polyethylene glycol linker, and each “—” is a covalent bond.

(105) Methods of using the composition for treatment of cell proliferative disorders are also within the invention. In an aspect of the invention, provided herein is a method of augmenting an anti-tumor immune response, including administering to a subject a composition comprising an immuno-stimulatory compound and a pHLIP® peptide.

(106) For example, the subject has a solid tumor. In other examples, the composition is injected directly into a tumor mass. Alternatively, the composition is systemically administered.

(107) As described herein, the immuno-stimulatory compound is delivered into the cytosols of cancer cells. Additionally, the composition as described herein is delivered into the cytosol of a macrophage within the tumor microenvironment.

(108) The immuno-stimulatory compound, as described herein, is delivered intracellularly to induce a biological effect. For example, the biological effect of the immuno-stimulatory compound is delivered in the presence of said pHLIP® is at least 20% greater than that delivered in the

absence of said pHLIP®.

(109) In some embodiments, the composition described herein targets the immune-stimulatory compound preferentially to a diseased tissue compared to a healthy tissue, thereby minimizing damage to said healthy tissue. In still other examples, the composition described herein selectively promotes intracellular delivery of the immuno-stimulatory compound to cells in diseased tissue. Moreover, the composition described herein selectively promotes intracellular delivery of the immuno-stimulatory compound into a cancer cell, or into macrophages in a tumor microenvironment, or into a macrophage in a diseased tissue environment.

(110) As provided herein, the methods of augmenting an anti-tumor immune response, including administering to a subject a composition comprising an immuno-stimulatory compound and a pHLIP® peptide, further include that the pHLIP® peptide comprises the amino acid sequence of Var3 pHLIP® ADDQNPWRAYLDLLFPTDTLLLDLLWCA (SEQ ID NO: 10), ADQDNPWRAYLDLLFPTDTLLLDLLWCA (SEQ ID NO: 212) or variations thereof.

(111) In some examples, the pHLIP® peptide comprises the a Var3 sequence with the amino acid sequence of AXDDQNPWRAYLDLLFPTDTLLLDLLWXA (SEQ ID NO: 2) or ADQDNPWRAYLDLLFPTDTLLLDLLWXA (SEQ ID NO: 214), wherein X is, selected from a lysine (Lys), a cysteine (Cys), or an Azido-containing amino acid.

(112) General Definitions

(113) Unless specifically defined otherwise, all technical and scientific terms used herein shall be taken to have the same meaning as commonly understood by one of ordinary skill in the art (e.g., in cell culture, molecular genetics, and biochemistry).

(114) As used herein, the term “about” in the context of a numerical value or range means $\pm 10\%$ of the numerical value or range recited or claimed, unless the context requires a more limited range.

(115) In the descriptions above and in the claims, phrases such as “at least one of” or “one or more of” may occur followed by a conjunctive list of elements or features. The term “and/or” may also occur in a list of two or more elements or features. Unless otherwise implicitly or explicitly contradicted by the context in which it is used, such a phrase is intended to mean any of the listed elements or features individually or any of the recited elements or features in combination with any of the other recited elements or features. For example, the phrases “at least one of A and B;” “one or more of A and B;” and “A and/or B” are each intended to mean “A alone, B alone, or A and B together.” A similar interpretation is also intended for lists including three or more items. For example, the phrases “at least one of A, B, and C;” “one or more of A, B, and C;” and “A, B, and/or C” are each intended to mean “A alone, B alone, C alone, A and B together, A and C together, B and C together, or A and B and C together.” In addition, use of the term “based on,” above and in the claims is intended to mean, “based at least in part on,” such that an unrecited feature or element is also permissible

(116) It is understood that where a parameter range is provided, all integers within that range, and tenths thereof, are also provided by the invention. For example, “0.2-5 mg” is a disclosure of 0.2 mg, 0.3 mg, 0.4 mg, 0.5 mg, 0.6 mg etc. up to and including 5.0 mg.

(117) A small molecule is a compound that is less than 2000 daltons in mass. The molecular mass of the small molecule is preferably less than 1000 daltons, more preferably less than 600 daltons, e.g., the compound is less than 500 daltons, 400 daltons, 300 daltons, 200 daltons, or 100 daltons.

(118) As used herein, an “isolated” or “purified” nucleic acid molecule, polynucleotide, polypeptide, or protein, is substantially free of other cellular material, or culture medium when produced by recombinant techniques, or chemical precursors or other chemicals when chemically synthesized. Purified compounds are at least 60% by weight (dry weight) the compound of interest. Preferably, the preparation is at least 75%, more preferably at least 90%, and most preferably at least 99%, by weight the compound of interest. For example, a purified compound is one that is at least 90%, 91%, 92%, 93%, 94%, 95%, 98%, 99%, or 100% (w/w) of the desired compound by weight. Purity is measured by any appropriate standard method, for example, by column

chromatography, thin layer chromatography, or high-performance liquid chromatography (HPLC) analysis. A purified or isolated polynucleotide (ribonucleic acid (RNA) or deoxyribonucleic acid (DNA)) or polypeptide is free of the amino acid sequences, or nucleic acid sequences that flank it in its naturally-occurring state. Purified also defines a degree of sterility that is safe for administration to a human subject, e.g., lacking infectious or toxic agents. A purified or isolated polynucleotide (ribonucleic acid (RNA) or deoxyribonucleic acid (DNA)) is free of the genes or sequences that flank it in its naturally-occurring state. A purified or isolated polypeptide is free of the amino acids or sequences that flank it in its naturally-occurring state.

(119) Similarly, by “substantially pure” is meant a nucleotide or polypeptide that has been separated from the components that naturally accompany it. Typically, the nucleotides and polypeptides are substantially pure when they are at least 60%, 70%, 80%, 90%, 95%, or even 99%, by weight, free from the proteins and naturally-occurring organic molecules with they are naturally associated.

(120) The transitional term “comprising,” which is synonymous with “including,” “containing,” or “characterized by,” is inclusive or open-ended and does not exclude additional, unrecited elements or method steps. By contrast, the transitional phrase “consisting of” excludes any element, step, or ingredient not specified in the claim. The transitional phrase “consisting essentially of” limits the scope of a claim to the specified materials or steps “and those that do not materially affect the basic and novel characteristic(s)” of the claimed invention.

(121) The terms “subject,” “patient,” “individual,” and the like as used herein are not intended to be limiting and can be generally interchanged. That is, an individual described as a “patient” does not necessarily have a given disease, but may be merely seeking medical advice.

(122) As used herein, the singular forms “a,” “an,” and “the” include the plural reference unless the context clearly dictates otherwise. Thus, for example, a reference to “a disease,” “a disease state”, or “a nucleic acid” is a reference to one or more such embodiments, and includes equivalents thereof known to those skilled in the art and so forth.

(123) As used herein, “treating” encompasses, e.g., inhibition, regression, or stasis of the progression of a disorder. Treating also encompasses the prevention or amelioration of any symptom or symptoms of the disorder. As used herein, “inhibition” of disease progression or a disease complication in a subject means preventing or reducing the disease progression and/or disease complication in the subject.

(124) As used herein, a “symptom” associated with a disorder includes any clinical or laboratory manifestation associated with the disorder, and is not limited to what the subject can feel or observe.

(125) As used herein, “effective” when referring to an amount of a therapeutic compound refers to the quantity of the compound that is sufficient to yield a desired therapeutic response without undue adverse side effects (such as toxicity, irritation, or allergic response) commensurate with a reasonable benefit/risk ratio when used in the manner of this disclosure.

(126) As used herein, “pharmaceutically acceptable” carrier or excipient refers to a carrier or excipient that is suitable for use with humans and/or animals without undue adverse side effects (such as toxicity, irritation, and allergic response) commensurate with a reasonable benefit/risk ratio. It can be, e.g., a pharmaceutically acceptable solvent, suspending agent or vehicle, for delivering the instant compounds to the subject.

(127) Examples are provided below to facilitate a more complete understanding of the invention. The following examples illustrate the exemplary modes of making and practicing the invention. However, the scope of the invention is not limited to specific embodiments disclosed in these Examples, which are for purposes of illustration only, since alternative methods can be utilized to obtain similar results.

(128) “Percentage of sequence identity” is determined by comparing two optimally aligned sequences over a comparison window, wherein the portion of the polynucleotide or polypeptide

sequence in the comparison window may comprise additions or deletions (i.e., gaps) as compared to the reference sequence (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage is calculated by determining the number of positions at which the identical nucleic acid base or amino acid residue occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison and multiplying the result by 100 to yield the percentage of sequence identity.

(129) The term “identical” or percent “identity,” in the context of two or more nucleic acids or polypeptide sequences, refer to two or more sequences or subsequences that are the same or have a specified percentage of amino acid residues or nucleotides that are the same (e.g., 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more identity over a specified region, e.g., of an entire polypeptide sequence or an individual domain thereof), when compared and aligned for maximum correspondence over a comparison window, or designated region as measured using a sequence comparison algorithm or by manual alignment and visual inspection. Such sequences that are at least about 80% identical are said to be “substantially identical.” In some embodiments, two sequences are 100% identical. In certain embodiments, two sequences are 100% identical over the entire length of one of the sequences (e.g., the shorter of the two sequences where the sequences have different lengths). In various embodiments, identity may refer to the complement of a test sequence. In some embodiments, the identity exists over a region that is at least about 10 to about 100, about 20 to about 75, about 30 to about 50 amino acids or nucleotides in length. In certain embodiments, the identity exists over a region that is at least about 50 amino acids in length, or more preferably over a region that is 100 to 500, 100 to 200, 150 to 200, 175 to 200, 175 to 225, 175 to 250, 200 to 225, 200 to 250 or more amino acids in length.

(130) For sequence comparison, typically one sequence acts as a reference sequence, to which test sequences are compared. In various embodiments, when using a sequence comparison algorithm, test and reference sequences are entered into a computer, subsequence coordinates are designated, if necessary, and sequence algorithm program parameters are designated. Preferably, default program parameters can be used, or alternative parameters can be designated. The sequence comparison algorithm then calculates the percent sequence identities for the test sequences relative to the reference sequence, based on the program parameters.

(131) A the “comparison window” refers to a segment of any one of the number of contiguous positions (e.g., least about 10 to about 100, about 20 to about 75, about 30 to about 50, 100 to 500, 100 to 200, 150 to 200, 175 to 200, 175 to 225, 175 to 250, 200 to 225, 200 to 250) in which a sequence may be compared to a reference sequence of the same number of contiguous positions after the two sequences are optimally aligned. In various embodiments, a comparison window is the entire length of one or both of two aligned sequences. In some embodiments, two sequences being compared comprise different lengths, and the comparison window is the entire length of the longer or the shorter of the two sequences. Methods of alignment of sequences for comparison are well-known in the art. Optimal alignment of sequences for comparison can be conducted, e.g., by the local homology algorithm of Smith & Waterman, *Adv. Appl. Math.* 2:482 (1981), by the homology alignment algorithm of Needleman & Wunsch, *J. Mol. Biol.* 48:443 (1970), by the search for similarity method of Pearson & Lipman, *Proc. Nat'l. Acad. Sci. USA* 85:2444 (1988), by computerized implementations of these algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group, 575 Science Dr., Madison, Wis.), or by manual alignment and visual inspection (see, e.g., *Current Protocols in Molecular Biology* (Ausubel et al., eds. 1995 supplement)).

(132) In various embodiments, an algorithm that is suitable for determining percent sequence identity and sequence similarity are the BLAST and BLAST 2.0 algorithms, which are described in Altschul et al., *Nuc. Acids Res.* 25:3389-3402 (1977) and Altschul et al., *J. Mol. Biol.* 215:403-410 (1990), respectively. BLAST and BLAST 2.0 may be used, with the parameters described herein, to

determine percent sequence identity for nucleic acids and proteins. Software for performing BLAST analyses is publicly available through the National Center for Biotechnology Information, as known in the art. This algorithm involves first identifying high scoring sequence pairs (HSPs) by identifying short words of length W in the query sequence, which either match or satisfy some positive-valued threshold score T when aligned with a word of the same length in a database sequence. T is referred to as the neighborhood word score threshold (Altschul et al., supra). These initial neighborhood word hits act as seeds for initiating searches to find longer HSPs containing them. The word hits are extended in both directions along each sequence for as far as the cumulative alignment score can be increased. Cumulative scores are calculated using, for nucleotide sequences, the parameters M (reward score for a pair of matching residues; always >0) and N (penalty score for mismatching residues; always <0). For amino acid sequences, a scoring matrix is used to calculate the cumulative score. Extension of the word hits in each direction are halted when: the cumulative alignment score falls off by the quantity X from its maximum achieved value; the cumulative score goes to zero or below, due to the accumulation of one or more negative-scoring residue alignments; or the end of either sequence is reached. The BLAST algorithm parameters W, T, and X determine the sensitivity and speed of the alignment. The BLASTN program (for nucleotide sequences) uses as defaults a wordlength (W) of 11, an expectation (E) of 10, M=5, N=-4 and a comparison of both strands. For amino acid sequences, the BLASTP program uses as defaults a wordlength of 3, and expectation (E) of 10, and the BLOSUM62 scoring matrix (see Henikoff & Henikoff, Proc. Natl. Acad. Sci. USA 89:10915 (1989)) alignments (B) of 50, expectation (E) of 10, M=5, N=-4, and a comparison of both strands.

OTHER EMBODIMENTS

(133) While the invention has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the invention, which is defined by the scope of the appended claims. Other aspects, advantages, and modifications are within the scope of the following claims.

(134) The patent and scientific literature referred to herein establishes the knowledge that is available to those with skill in the art. All United States patents and published or unpublished United States patent applications cited herein are incorporated by reference. All published foreign patents and patent applications cited herein are hereby incorporated by reference. Genbank and NCBI submissions indicated by accession number cited herein are hereby incorporated by reference. All other published references, documents, manuscripts and scientific literature cited herein are hereby incorporated by reference.

(135) While this invention has been particularly shown and described with references to preferred embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the invention encompassed by the appended claims.

Claims

1. A composition comprising an immuno-stimulatory compound (ISC) and a pH low insertion peptide (pHLIP) peptide, wherein said ISC is selected from a group consisting of: (i) a cyclic dinucleotide (CDN), (ii) a cyclic purine dinucleotide, (iii) a cyclic purine dinucleotide which binds to stimulator of interferon genes (STING), (iv) a cGAMP, (v) a 3',5'-cyclic diadenylic acid (c-di-AMP), and (vi) a cyclic diguanylate (c-di-GMP) cyclic compound.
2. The composition of claim 1, comprising the following structure: Peptide-Linker-ISC wherein "Peptide" is a pHLIP peptide comprising the sequence TABLE-US-00008 (SEQ ID NO: 212) ADQDNPWRAYLDLLFPTDTLLLDLLWCA or (SEQ ID NO: 10) ADDQNPWRAYLDLLFPTDTLLLDLLWCA or (SEQ ID NO: 213) ADQDNPWRAYLDLLFPTDTLLLDLLWKA or (SEQ ID NO: 217)

ADDQNPWRAYLDLLFPTDTLLLDLLWKA wherein "Linker" is a cleavable linker, wherein "ISC" is a cyclic purine dinucleotide which binds to stimulator of interferon genes (STING), and wherein each "—" is a covalent bond.

3. The composition of claim 2, wherein said linker comprises a disulfide bond or an acid-labile bond.

4. The composition of claim 2, wherein said linker is self-immolating.

5. The composition of claim 1, further comprising a modulator of polarity.

6. The composition of claim 1, wherein said composition comprises 2 or more pHLIP peptides.

7. The composition of claim 6, comprising the following structure: Peptide1-Link-Peptide2

wherein "Peptide1" is a first pHLIP peptide comprising the sequence TABLE-US-00009 (SEQ ID NO: 1) ADDQNPWRAYLDLLFPTDTLLLDLLWXA or (SEQ ID NO: 214)

ADQDNPWRAYLDLLFPTDTLLLDLLWXA, wherein "Peptide2" is a second pHLIP peptide comprising the sequence TABLE-US-00010 (SEQ ID NO: 1)

ADDQNPWRAYLDLLFPTDTLLLDLLWXA or (SEQ ID NO: 214)

ADQDNPWRAYLDLLFPTDTLLLDLLWXA, wherein "X" indicates any amino acid residue, including a lysine (Lys), a cysteine (Cys), or an Azido-containing amino acid, wherein "Link" is a polyethylene glycol linker, and wherein each "—" is a covalent bond.

8. A method of augmenting an anti-tumor immune response, comprising administering to a subject a composition comprising an immuno-stimulatory compound and a pHLIP peptide, wherein said immuno-stimulatory compound is selected from the group consisting of: (i) a cyclic dinucleotide (CDN), (ii) a cyclic purine dinucleotide, (iii) a cyclic purine dinucleotide which binds to stimulator of interferon genes (STING), (iv) a cGAMP, (v) a 3',5'-cyclic diadenylic acid (c-di-AMP), and (vi) a cyclic diguanylate (c-di-GMP) cyclic compound.

9. The method of claim 8, wherein said subject comprises a solid tumor.

10. The method of claim 8, wherein said composition is injected directly into a tumor mass.

11. The method of claim 8, wherein said composition is systemically administered.

12. The method of claim 8, wherein said immuno-stimulatory compound is delivered into the cytosols of cancer cells.

13. The method of claim 8, wherein said immuno-stimulatory compound is delivered into the cytosol of a macrophage within the tumor microenvironment.

14. The method of claim 8, wherein said immuno-stimulatory compound is delivered intracellularly to induce a biological effect.

15. The method of claim 14, wherein the biological effect of said immuno-stimulatory compound delivered in the presence of said pHLIP is at least 20% greater than that delivered in the absence of said pHLIP.

16. The method of claim 8, wherein said composition targets said immuno-stimulatory compound preferentially to a diseased tissue compared to a healthy tissue, thereby minimizing damage to said healthy tissue.

17. The method of claim 8, wherein said composition selectively promotes intracellular delivery of said immuno-stimulatory compound to cells in diseased tissue.

18. The method of claim 17, wherein said composition selectively promotes intracellular delivery of said immuno-stimulatory compound into a cancer cell.

19. The method of claim 8, wherein said composition selectively promotes intracellular delivery of said immuno-stimulatory compound into macrophages in a tumor microenvironment.

20. The method of claim 8, wherein said composition selectively promotes intracellular delivery of said immuno-stimulatory compound into a macrophage in a diseased tissue environment.

21. The method of claim 8, wherein said pHLIP peptide comprises the amino acid sequence of TABLE-US-00011 (SEQ ID NO: 10) ADDQNPWRAYLDLLFPTDTLLLDLLWCA or (SEQ ID NO: 212) ADQDNPWRAYLDLLFPTDTLLLDLLWCA.

22. The method of claim 8, wherein the composition comprises the following structure: Peptide-

Link-ISC wherein "Peptide" is a pHLIP peptide comprising an amino acid sequence of TABLE-US-00012 (SEQ ID NO: 2) AXDDQNPWRAYLDLLFPTDTLLLDLLWXA or (SEQ ID NO: 214) ADQDNPWRAYLDLLFPTDTLLLDLLWXA, wherein "X" is, selected from a lysine (Lys), a cysteine (Cys), or an Azido-containing amino acid, wherein "Link" is a polyethylene glycol linker, wherein "ISC" is said immuno-stimulatory compound, and wherein each "—" is a covalent bond.
